

A detailed pencil sketch of three skulls and a rose. The skulls are arranged in a triangular pattern, with one at the top left, one at the bottom left, and one on the right. The top-left skull is shown in profile, facing right. The bottom-left skull is shown from a front-three-quarter view, facing right. The skull on the right is shown from a front-three-quarter view, facing left. A rose is sketched on the left side, partially overlapping the top-left skull. The background features a dense, dark, leafy pattern at the top, and a lighter, textured area below it. The overall style is a detailed pencil drawing.

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SAPIENT

The Undergraduate Journal of
Biological Anthropology

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Spring 2025**

Sapient is Columbia University's peer-reviewed undergraduate journal of biological anthropology. Founded in 2012 to give undergraduates at Columbia a place to share and reflect upon bioanthropological works of academic and artistic merit, *Sapient* has continued to develop. For questions, comments, or to get involved visit sapientjournal.com.

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Letter from the Editors

Dear Reader,

On behalf of the Editorial Board, we are proud to present the 12th Volume of *Sapient*, Columbia University's Undergraduate Journal of Biological Anthropology. This year's volume reflects the vastness that exists in biological anthropology, with a wide range of paper topics, along with assorted visual artwork and creative writing. We hope you find this volume as entertaining and intriguing as we did.

This year, we were impressed our board's drive and commitment to make more than a publication, but a strong community with a shared interest in biological anthropology. We are incredibly grateful for the relationships we have cultivated this year in our board and with students across campus. We were thrilled to host several events where we got to meet with students interested in biological anthropology and introduce them to our journal and a space to feed that interest. Our events and this journal could not have been possible without the guidance and support of our advisor, Dr. Jill Shapiro.

Both of us will graduate this year, leaving the journal in the capable hands of two new Co-Editors-in-Chief. We believe that *Sapient* has the power to bond students over their shared interests, and we have been made better for it.

Rahele Megosha and Lily Kuhn, Co-Editors-in-Chief

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Artwork by Kah Sin Tang

Ode to Orangutan

By Kah Sin Tang

Verse 1]

Swingin' through the jungle, crown tilted just so,
Orange flame in the branches, let the low winds
blow.

Fingers stretch long like they're scribblin' laws,
Built the monarchy, never took a single pause.

[Verse 2]

Swingin' in the jungle, kingpin of the trees,
Big orange vibe, makin' lemurs freeze.
Loris tryin' to keep up, movin' all slow,
Tarsier wide-eyed, but they not in the show.

[Chorus]

King of the trees, don't challenge the rule,
Lemurs hang back 'cause they ain't got the tools.
Loris too slow, tarsier's eyes too wide,
Chimps talk big but they can't match the stride.

[Verse 3]

Chimp got the brain, but I got the style,
Monkeys tryna mimic, but that takes a while.
From Borneo to Sumatra, I'm the G.O.A.T.,
Swingin' to the beat, they just rockin' the boat.

[Verse 4]

Monkeys throwin' shade but their tails just weak,
Gorillas pound chests but their grip can't speak.
Hands flex wisdom, ancient vine slang,
I'm the orangutan, call me the Orangut-King.

[Chorus]

King of the trees, don't test the decree,
Swing sharper than axes, roots bow to me.
From canopy to canopy, shadows I command,

Every beast in the jungle knows the feel of my
hand.

[Bridge]

Citrus sunrise coat, blaze in the green,
Golden hour glow, I'm the jungle's sheen.
But aye aye, hold up—
no no we don't talk about the aye aye
don't mention the cursed,
Name unspoken while the rest disperse.

[Verse 5]

Ginger fur gleamin', sunbeam sway,
Arms outstretched, I glide all day.

Other primates hatin', they stuck on the ground,
While I'm up here with a crown profound.

[Verse 6]

I see lemurs schemin', always lookin' overdue,
Loris movin' stealth, thinkin' slick, not true.
Tarsier's buggin' out, eyes wide like moons,
Gorilla stompin', no rhythm to their tunes.

[Verse 7]

Bananas in my pocket, wisdom in my head,
Leaf umbrella poppin' if it rains instead.
Slow loris says "bruhhh," I just laugh back,
King orangutan never off-track.

[Final Chorus]

But we don't talk about aye aye, aye no-no,
Eerie fingers creepin', movin' so slow.
Vibes too strange, won't mess with the flow,
Aye aye keepin' secrets we don't wanna know!

Fear-of-Childbirth, Hypnobirthing, and Offspring Development: An Evolutionary Explanation for Hypnotizability

M. Hartog, Columbia University

Introduction

Hypnotizability is a person's tendency to become absorbed in their internal world, focus deeply on a specific experience, and detach from other stimuli. Prominent scholars have proposed that high hypnotizability is an evolutionary adaptation (Spiegel, 2008; Winkelman, 2013; Krippner *et al.*, 2019). Spiegel (2008), for instance, theorizes that highly hypnotizable hominins were more socially attuned and avoided predation more effectively, while Krippner *et al.* (2019) posit that those with higher hypnotic capacity were more responsive to shamanic healing rituals and could subsequently better fight off infectious diseases.

Over the past few decades, an alternative type of birthing program has gained traction in the United States, prompting another hypothesis for the evolution of hypnotizability. Popularized by the likes of Hollywood stars Angelina Jolie and Jessica Alba, as well as Kate Middleton, Duchess of Cambridge, 'hypnobirthing' teaches prospective mothers self-hypnosis to mitigate pregnancy-related anxiety and enable a smoother birthing process (Madden *et al.*, 2016; Goad, 2021).

The vast majority of studies on antenatal self-hypnosis methods focus on maternal experiences and showcase a reduced anxiety response in participants (Catsaros and Wendland, 2023). However, anxiety during pregnancy has also

been linked to a range of adverse developmental outcomes and health issues in offspring, including low birth weight (LBW), reduced likelihood of rapid weight gain (RWG) during the initial postnatal years, and cardiovascular, metabolic, immune, and psychological conditions later in life (Caparros-Gonzalez *et al.*, 2021; Zhou *et al.*, 2023). The ability to reduce this type of anxiety should thus benefit not only the mother's mental health, but also her child's well-being. Individuals with naturally high hypnotic capacity may auto-induce the same types of altered states of consciousness (ASCs) that hypnobirthing programs aim to evoke. This paper thus examines whether the trait of hypnotizability evolved to aid the process of labor and promote offspring development.

Background

Hypnosis and Hypnotizability

'Hypnosis' and 'hypnotizability' are related, yet distinct terms. Hypnosis refers to a process in which individuals enter an ASC, intensely focus on an action or experience, and reduce their awareness of additional context. This allows for shifts in their perception, meaning, and cognition (Spiegel, 2008). Shamanic healers, clinical hypnotists, and other outside parties can guide humans into ASCs, but humans may also auto-induce such states (McClenon, 1997). The

trait of hypnotizability describes a person's innate capacity to do so and subsequently respond to suggestion, whether through prompting from external sources or internal initiation (Spiegel, 2008; Dell, 2021; Elkins, 2021).

Most people are hypnotizable to some extent, but individuals with inherently high hypnotizability are more inclined to respond to hypnosis techniques and interventions (Elkins, 2021). Numerous assessments such as the Hypnotic Induction Profile (HIP), the Harvard Group Scale of Hypnotic Susceptibility (HGSHS), the Elkins Hypnotizability Scale (EHS) (Elkins, 2021), and the Stanford Hypnotic Susceptibility Scale (SHSS) measure the trait (Woody *et al.*, 2005). Around 10-15% of the population score highly on these scales (Woody *et al.*, 2005) and longitudinal studies show that hypnotizability remains stable over a 25-year period (Piccione *et al.*, 1989).

Antenatal Anxiety and Hypnobirthing

Hypnobirthing is based on the premise that reducing mothers' fear of labor will lower their anxiety levels, which in turn will reduce their experienced pain during the actual event (Beebe, 2014). This, in theory, works along both physiological and psychological pathways. Excessive fear can aggravate labor pain due to increased muscular tension (Madden *et al.*, 2016) and stimulate catecholamine secretion, which intensifies pain perception and prompts muscle spasms and vasoconstriction (Kamalimanesh *et al.*, 2025). Undeniably, cervical dilatation and

uterine contractions evoke bodily discomfort, but sociocultural factors also play a role in how pain is anticipated and experienced, as individuals' interpretation of pain often aligns with their learned beliefs and behavior (Lowe, 2002). These are deemed alterable through (self-)hypnosis.

Hypnobirthing typically starts by instructing pregnant individuals to linguistically reframe the birthing experience from painful and negative to peaceful and powerful—together with body awareness, introspection, and visualization exercises, the use of repetitive positive language is intended to alter the perception of pain. The purpose of these methods is to create a conditioned response, so that when labor begins, stress hormone activity is reduced and uterine contractions progress more efficiently (Beebe, 2014). Then, 'deepening techniques' are introduced to enable hypnoanalgesia: through practice and conditioning, birth nociception (pain sensations) is reimagined as numbness or mentally reattributed to another body part (Beebe, 2014).

Ultimately, the aim of hypnobirthing practices is to prepare pregnant individuals for labor and make that experience less distressing by teaching how to induce an altered state of mind—similar to states that highly hypnotizable humans can enter spontaneously (Spiegel, 2008). Moreover, research on antenatal self-hypnosis shows it eases anxiety during pregnancy: Nisa *et al.* (2024) found that women who engaged in the practice had significantly lower anxiety scores on the Depression, Anxiety, and Stress Scale (DASS).

Kamalimanesh *et al.* (2025) and Seitz *et al.* (2024) reiterate this finding, noting reduced anxiety and fear-of-childbirth, decreased need for surgical birth intervenance, and lower rates of gestational hypertension. A literature review by Catsaros and Wendland (2023) concluded that hypnosis during pregnancy improves mothers' overall emotional stability and potentially positively impacts their psychological well-being after birth.

Implications for Offspring

Most studies on antenatal hypnosis focus on mothers' experiences (Madden *et al.*, 2016; Catsaros and Wendland, 2023; Nisa *et al.*, 2024; Seitz *et al.*, 2024; Kamalimanesh *et al.*, 2025), and few directly explore how it impacts offspring. However, it is plausible that improvements in child health outcomes correspond with how effective the practice is in reducing mothers' anxiety.

Anxiety during pregnancy can raise mothers' cortisol levels, and increase the amount of cortisol transmitted to the fetus via the placenta—elevated glucocorticoids (GCs) in the placenta restrict fetal growth and are associated with preeclampsia or preterm births to infants with low birth weights (Caparros-Gonzalez *et al.*, 2021). Maternal anxiety during the second and third trimester is also correlated with reduced likelihood of rapid weight gain (RWG) in the infant's first years of life and lower weight and Body Mass Index (BMI) by preschool age (Zhou *et al.*, 2023). Furthermore, placental cortisol and serotonin transmissions can cause long-term

changes to a fetus's hypothalamic-pituitary-adrenal (HPA) axis, increasing their vulnerability to later-in-life health concerns such as obesity, hypertension, asthma, eczema, affective disorders, schizophrenia, depressive and anxiety disorders, and cerebral palsy (Caparros-Gonzalez *et al.*, 2021).

Discussion

Hypnobirthing methods rely on the logic that fear of birthing triggers antenatal anxiety and that self-hypnosis prompts a reduction of this anxiety, in turn minimizing the perception of pain during labor (Kamalimanesh *et al.*, 2025). More importantly, a reduction of antenatal anxiety likely benefits offspring by supporting fetal and postnatal growth and decreasing the risk of later-life diseases and disorders (Caparros-Gonzalez *et al.*, 2021; Zhou *et al.*, 2023).

A similar chain of reasoning may apply to naturally high hypnotizability. Spiegel (2008) asserts that individuals with this trait are intensely imaginative, can focus deeply on critical tasks, and live in the moment. These characteristics enable them to manage pain or fear well, and from an evolutionary perspective would have helped early hominins remain calm and collected in the face of predation. Extending his ideas, mothers with high hypnotic capacity would be inherently better equipped to tune out pregnancy-related fears, thus facilitating an intrauterine environment with reduced exposure to maternal stress hormones.

McClenon (1997) suggests that genes for

high hypnotizability were selected for, as *Homo sapiens* with this trait benefited more from shamanic healing rituals—practices that induced ASCs and yielded favorable psychological and physiological effects. Krippner *et al.* (2019) add that these healing ceremonies were particularly useful in times before antibiotics were available to treat diseases, and that highly hypnotizable individuals were more inclined to respond to them. Similarly, this paper proposes that before epidurals and analgesics were readily available to ease pain during childbirth, humans with high hypnotizability may have better managed the stress and pain associated with labor than their less hypnotizable counterparts—either through a healer’s suggestion or their natural ability to tune out pain.

Winkelman (2013) theorizes that shaman-induced altered states of consciousness allowed Paleolithic humans to experience new ways of perceiving both themselves and others, leading to increased empathy and social awareness. Resultantly, societies became more cooperative, with members increasingly contributing to group care and food security. This raises the possibility that social dynamics may have reduced prenatal stressors; if mothers did not have to worry about food procurement during pregnancy and could rely on support during labor, their anxiety levels—and subsequently, the fetal exposure to maternal stress hormones—would have been greatly reduced.

Contemporary research on hypnosis practice during pregnancy shows almost universally

reduced anxiety amongst mothers, although the degree of this reduction varies across studies (Madden *et al.*, 2016; Catsaros and Wendland, 2023). One explanation for this variance is differences in practice frequency, as Beebe (2014) posits that the efficacy of self-hypnosis on childbirth is dependent on consistency and the amount of training.

Differences in rapport between birthing instructors and mothers, birthing location, and external stressors during labor can affect mothers’ comfortability (Chen *et al.*, 2023). Birthing classes have also increasingly been offered virtually, which may or may not provide the same level of relational or environmental safety as in-person classes (Mousavi *et al.*, 2021), and this presumably affects individuals’ ability to enter altered states. Such variations in research protocol may limit the comparability of findings. Although each study discussed in this paper standardized its own methods, (self-) hypnosis practices were not uniform across studies (Madden *et al.*, 2016; Catsaros and Wendland, 2023; Nisa *et al.*, 2024; Seitz *et al.*, 2024; Kamalimanesh *et al.*, 2025). Future research could harmonize session frequency, duration, and setting, to more accurately gauge efficacy.

The discrepancy in anxiety reduction across studies might also be explained by an uneven distribution of highly hypnotizable pregnant individuals across samples. Elkins (2021) emphasizes that nearly everyone can benefit from hypnosis interventions, but those who are highly hypnotizable will respond more quickly and

reap greater benefits from the practices. If some studies inadvertently included larger samples of highly hypnotizable individuals, this could influence the results.

It appears evident that hypnotizability plays a role in mitigating anxiety during pregnancy and improving birth outcomes. High hypnotic capacity, therefore, could be advantageous not just for an individual, but for populations at large. Despite this, only 10-15% of the population is highly hypnotizable (Woody *et al.*, 2005). One evolutionary explanation for why this percentage is not higher is that individuals with the trait may not recognize important pregnancy-related cues like pain or hunger, which could be detrimental to both their survival and that of their offspring.

Summary and Conclusion

This paper explored whether the trait of hypnotizability evolved to improve labor and offspring outcomes by linking the physiological mechanisms of antenatal anxiety to child health, and the efficacy of contemporary birthing techniques such as hypnobirthing. Mothers' anxiety during pregnancy can severely impact fetal and postnatal health, as stress hormones restrict fetal growth, cause preterm births, and permanently modify the fetus' HPA axis, making them vulnerable to a slew of long-term disorders (Caparros-Gonzalez *et al.*, 2021). The ability to reduce this anxiety therefore seems evolutionarily favorable.

While hypnobirthing itself is focused on bettering mothers' pregnancy and birthing

experiences (Beebe, 2014), the methods employed through these practices should, in theory, positively impact the prospective child as well. It stands that individuals who can spontaneously auto-hypnotize themselves in the way that hypnobirthing teaches would reap similar benefits.

Other evolutionary explanations exist for why hypnotizability emerged as a trait in the human lineage (McClenon, 1997; Winkelman, 2013; Krippner *et al.*, 2019), but the theory that it facilitated less-stressful pregnancies and labors and subsequently promoted healthier offspring is not mutually exclusive—it appears complementary. Archeological evidence of shamanic rituals specifically related to labor practices could strengthen this hypothesis.

It should be noted that high hypnotizability may come with certain disadvantages (Spiegel, 2008). Individuals with this trait might dissociate from pain or bodily signals so effectively that they disregard pregnancy-related cues of their baby being in distress or warning signs of their own health. The benefits of reducing fear-of-childbirth and anxiety during pregnancy might still outweigh these risks.

More research needs to be done to determine the exact pathways through which hypnotizability and self-hypnosis impact reproductive success. Longitudinal studies that track the development of children born to mothers who practiced self-hypnosis during pregnancy might offer interesting insight. Overall, there is a case to explore an expanded role

for (self-)hypnosis techniques in biomedicine. Specifically in obstetrics, such approaches could have tremendous implications for both maternal mental health and offspring health outcomes. Current self-hypnosis-based anxiety reduction methods are considered holistic alternatives or additions to basic prenatal care, and are relatively costly. The total course fee of one of the most popular self-hypnosis franchises, HypnoBirthing, can cost up to \$300-600 for five classes, location dependent (“HypnoBirthing Classes,” n.d.). This cost would make program participation unattainable for many women of lower socioeconomic status, who have increased likelihood to experience high stress environments (Verbeek *et al.*, 2019). These individuals may have the greatest need for support in managing anxiety. Offering self-hypnosis training in mainstream medical settings or in low-cost clinics could significantly benefit the emotional well-being of marginalized individuals and the health of their offspring. Since these types of programs do not require medical equipment, they could also be offered in community centers, psychological and counseling practices, or made more accessible online. For the portion of the population with high hypnotizability, such practices likely will prove especially beneficial.

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A Proud Little Talapoin

By Tatem Kegley

A proud little talapoin once heard,
“Is that a squirrel on that branch or a bird?”
So he jumped from his cache,
In the water he splashed,
And heard, “My! What a fish!” from the shore.

The Mandrill

By Sarah Morris

We learned to be wary of spandrels
But have you considered the mandrill?
Evolved to perfection,
And if given election,
How well this monkey would handle.

Painted in red and in blue
And a beard of clementine hue,
This terrestrial runner
Is a capital stunner,
The ideal for me and for you!

Ladders to Nowhere, Slightly Hairless, Fully Lost

By Kah Sin Tang

Here we are,
standing on a ladder we built ourselves,
assembled from thought, from invention, from fear,
peering down at the world below,
as slightly hairless apes who once clung to branches,
now clinging instead to ideas of difference.

We scan the Earth,
this pale stone adrift in a vast and ancient dark,
seeking endlessly the boundaries between “us” and “them,”
between “self” and “other,”
as if carving lines into water could make it hold shape.

We name, we label, we divide:
species, color, culture, creed,
desperate to prove that somewhere, somehow,
We are singular in essence and set apart in meaning.

It is a tender defiance of the truth
written into every atom of our being:
that we are stitched from the same cloth,
born of the same dust that spins the stars,
woven of atoms and quarks
that care nothing for the stories we tell.

The hydrogen that kindled the first suns
still courses through our blood.
The calcium that binds our bones
once drifted in forgotten seas.
The iron that sparks in our veins
was forged in the death-throes of ancient stars,
before Earth was even a dream.

And yet we forget.
We forget that every border we draw is drawn within ourselves.
We forget that difference is illusion,
that uniqueness is not a fortress,
but a fleeting ripple across a shared sea.

In our terror of sameness,
we build ladders taller and taller,
scrambling upward to declare our dominion,
only to find that the higher we climb,
the clearer it becomes:
We are all the same breath,
the same song,
the same trembling hope
wearing different faces.
Perhaps one day,
we will step down from the ladder,
kneel in the dust,
and feel the pulse of the universe beneath our fingertips,
not above, not apart,
but as ripples in the same infinite sea.



Artwork by Kah Sin Tang

Evolutionary Roots of Homosexuality in a Cooperative Breeding Context

Adib Rabbani, Columbia University

“Among the most social of all mammals, humans survived not because they were the strongest, but because they nurtured, bonded, and cooperated.”

– Sarah Blaffer Hrdy, *Mothers and Others*

In evolutionary biology, traits that persist across generations typically offer some reproductive or survival benefit (Hamilton, 1964). Homosexual behavior, which often leads to fewer or no direct offspring, seems puzzling in this context. One hypothesis suggests that homosexuality offers indirect fitness advantages through alloparenting, or caregiving by individuals other than the biological parents (Hrdy, 2009; Hamilton, 1964). In many species, non-reproductive individuals help raise young, promoting kin survival and shared genes

(Hamilton, 1964). The “gay uncle/aunt” hypothesis similarly holds that homosexual individuals can increase relatives’ reproductive success by helping to raise their kin (Kirkpatrick, 2000; Bobrow and Bailey, 2001). Researchers argue that homosexual behavior persists in human and non-human populations due to its role in kin support and cooperative child-rearing (Kirkpatrick, 2000; Bobrow and Bailey, 2001; Hrdy, 2009). I argue that homosexual behavior persists across species not as a paradox, but as part of the cooperative breeding strategies that have long defined human evolution. This essay explores how these dynamics operate—focusing

on alloparenting, genetic factors, and animal behavior.

How Alloparents Support Survival

Alloparenting is common in cooperative breeders, where kin assist a breeding pair in raising offspring (Lee, 1987; Hrdy, 2009). These helpers feed, carry, protect, and teach the young (Lee, 1987). For example, juvenile female elephants babysit their calves while their mothers forage, improving calf survival and conserving maternal energy (Lee, 1987). In primates like marmosets and tamarins, older siblings help care for their younger siblings, thereby shortening inter-birth intervals (Hrdy, 2009). In addition, chimpanzees display alloparental care on occasion (Hrdy, 2009). Humans exemplify this strategy—child-rearing has long included grandparents, siblings, and extended kin. Sarah Hrdy emphasizes that human mothers evolved in a ‘it took a village’ context to raise a child, with alloparents enabling faster reproductive success (Hrdy, 2009).

Cooperative breeding enhances offspring survival and reduces maternal burden (Hrdy, 2009). In species with alloparents, young often grow faster and healthier (Hrdy Mothers and Others 211). For example, elephant calves with more helpers are better protected and nourished (Lee, 1987). Helpers also gain indirectly through kin selection—helping relatives propagate

shared genes (Hamilton, 1964). Assisting kin can lead to evolutionary success, even without reproducing (Hamilton, 1964). This supports how traits like homosexuality might persist if they promote family reproductive success. In cooperative breeding systems, altruistic behaviors are selected for when beneficiaries are related (Kirkpatrick, 2000). Homosexual individuals, not raising their own children, may have historically been well-suited to become such advantageous alloparents.

The Genetic Foundations of Homosexuality

To understand homosexual alloparenting's evolutionary role, it is first necessary to establish that same-sex orientation has a biological basis. One large population study of Swedish twins found that genetic factors account for 34–39% of variation in male same-sex behavior, with higher concordance in identical to fraternal twins (Långström *et al.*, 2010).

Despite reducing direct reproduction, genes linked to homosexuality persist at stable rates (Camperio-Ciani *et al.*, 2004), implying compensatory advantages—a case of balancing selection.

One explanation is sexually antagonistic selection, where certain alleles benefit one sex while disadvantaging the other (Camperio-Ciani *et al.*, 2004). Camperio-Ciani *et al.* (2004) found that mothers and maternal aunts of gay men had significantly higher fertility, suggesting a maternally inherited factor—possibly X-linked—that boosts female reproduction while increasing

male homosexuality (Camperio-Ciani *et al.*, 2004). This trade-off may favor gene persistence at the family level.

Another proposed mechanism links sexual orientation to personality profiles—lower aggression and higher empathy—that may foster caregiving and social cohesion (Gladue and Bailey, 1995; Barron and Hare, 2019). These traits could enhance caregiving and group cohesion (Barron and Hare, 2019; Hrdy, 2009). Evolutionary psychologists Andrew Barron and Brian Hare argue that same-sex attraction may have evolved as part of a “self-domestication” process favoring reduced aggression and increased social bonding (Barron and Hare, 2019). In this view, homosexuality could reflect a broader genetic disposition toward nurturing and alliance-building (Barron and Hare, 2019). These traits are highly valuable in a cooperative species (Hrdy, 2009).

Nonetheless, some scholars view same-sex orientation as not directly selected but potentially arising from other selected traits. Some researchers propose that homosexuality is not adaptive in itself, but a byproduct of other advantageous traits such as high sensitivity or creativity (Kirkpatrick, 2000; Bobrow and Bailey, 2001). While the byproduct hypothesis may hold in some contexts, cross-species and cross-cultural patterns of homosexual caregiving suggest a functional contribution to kin support (de Waal, 1995; Vasey and VanderLaan, 2010; Nila *et al.*, 2018). The kin-selection and cooperative-breeding framework offers a more direct

evolutionary explanation—one grounded in the inclusive-fitness benefits of alloparental behavior (Hamilton, 1964; Hrdy, 2009).

Together, these theories suggest homosexuality is not maladaptive but tied to traits that support kin and community. Genes linked to same-sex orientation may persist because they foster cooperative behavior or enhance female fertility—routes through which they contribute to evolutionary success.

Animal Models of Same-Sex Kinship

Same-sex behavior and cooperative parenting often coexist across the animal kingdom (de Waal, 1995; Hrdy, 2009). Biologists have documented homosexual interactions in hundreds of species, frequently linked to social bonding and caregiving (de Waal, 1995; Vasey and VanderLaan, 2010; Nila *et al.*, 2018). A notable example is the bonobo (*Pan paniscus*), a matriarchal ape known for using sex as social glue (de Waal, 1995). Bonobos engage in frequent sexual behaviors between both same and opposite sexes (de Waal, 1995). Female bonobos form strong alliances through genital rubbing, enabling them to dominate males and maintain group peace (de Waal, 1995). These bonds foster trust and solidarity; newly immigrated females often engage in same-sex play with resident females to integrate into the group (de Waal, 1995). Additionally, these alliances benefit offspring—female-led groups are safer and mothers often co-feed and jointly protect each other's infants.

Laysan albatrosses (*Phoebastria immutabilis*)

on Oahu offer another example of same-sex pairing linked to cooperative parenting (Young and VanderWerf, 2014). In one colony, 31% of pairs were female–female, forming long-term bonds to cooperatively raise chicks. Though female–female pairs fledged fewer chicks than heterosexual pairs, they still produced more offspring than they would have alone. In this context, “fledging” refers to the successful raising of a chick until it leaves the nest, a measure of reproductive success widely used in ornithology (Young and VanderWerf, 2014). Their pairing reflects a flexible, evolutionarily viable strategy under demographic pressure.

Male–male pairs have been observed nesting and raising chicks in penguin species, especially in captivity (Ellis *et al.*, 2002). This example shows that the drive to nurture can exist outside heterosexual mating. Beyond birds, many species illustrate how cooperative and homosexual behaviors intertwine to support kinship and survival. Among insects like ants and bees, sterile workers devote their lives to raising the queen's offspring—an extreme form of inclusive fitness (Hamilton, 1964). While not homosexuality, these systems reinforce the evolutionary power of altruistic care over reproduction (Hrdy, 2009). Similarly, homosexual behavior in vertebrates often emerges in cooperative social structures, supporting alliances and caregiving (de Waal, 1995; Vasey and VanderLaan, 2010).

From bonobos to albatrosses, the animal kingdom has many examples of same-sex behavior enhancing survival and social stability

(de Waal, 1995; Young and VanderWerf, 2014). These patterns demonstrate that such behavior serves evolutionary functions—far from unnatural, it is part of nature’s adaptive toolkit.

Why Evolution Favors the Helpers

The persistence of homosexual behavior can be explained by evolutionary advantages, particularly in a cooperative breeding environment. A primary factor is inclusive fitness: by assisting in the care of close relatives, homosexual individuals enhance the survival and ultimate reproduction of kin who share their genes (Hamilton, 1964; Hrdy, 2009). In ancestral human environments with high child mortality and limited maternal resources, an extra caregiver—like a childless aunt or uncle—could make the critical difference for a child’s survival (Hrdy, 2009). Genes that predisposed individuals to this helper role may have been favored despite reduced personal fertility. Studies of Samoan *fa’afafine*—male-at-birth individuals who take on a feminine gender role—support this idea: their intensive caregiving improves their nieces’ and nephews’ outcomes (Vasey and VanderLaan, 2010). Even if kin-directed altruism doesn’t fully offset reproductive costs, it contributes significantly over evolutionary time. One Indonesian study estimated that such care compensated for about 20% of reproductive loss—enough for the genes involved to persist (Nila *et al.*, 2018). In harsh conditions, the presence of devoted homosexual alloparents may have tipped the scales between kin survival and death.

Critics may point out that not all homosexual individuals engage in kin-directed altruism, and in many modern societies, cultural barriers may prevent them from doing so. This is a valid observation—just as not every heterosexual individual becomes a parent, not every homosexual person serves as an alloparent. However, natural selection acts on average tendencies, not universal behaviors. The consistent presence of elevated avuncular care in cultures that affirm third-gender roles (like the *fa’afafine* or *hijras*) suggests that, when supported by social norms, homosexual individuals are especially likely to invest in kin (Vasey and VanderLaan, 2010; Nanda, 1990; Williams, 2010). This pattern of caregiving, even if not universal, is evolutionarily meaningful.

Beyond kin-directed altruism, another advantage is reduced intrasexual conflict. In species with fierce male competition for mates, a subset of non-competitive males—like homosexual individuals—can lower reproductive tension (Gladue and Bailey, 1995). In early human groups, a gay male might bond with others and contribute to communal tasks without entering mating rivalries, promoting group cohesion. Populations where 5–10% of males didn’t compete for mates may have had less conflict and greater male survival (Kirkpatrick, 2000). Research shows that homosexual men often display lower hostile masculinity and aggression (Gladue and Bailey, 1995), suggesting they may have been natural mediators or peacekeepers. Bonobos show a similar dynamic: same-sex interactions

help diffuse aggression, sustaining troop harmony (de Waal, 1995). It's plausible that early human groups similarly benefited from these peace-promoting roles.

An additional evolutionary benefit may stem from elevated empathy and prosocial behavior. Studies have found homosexual individuals, on average, to test more highly for empathy and caretaking (Gladue and Bailey, 1995). These findings come primarily from studies of homosexual males. Rather than being ubiquitous, these traits can be part of the extended genetic toolkit of same-sex orientation (Hrdy, 2009). In cooperative breeding schemes, emotional intelligence and altruism prove especially valuable—caring for the young, helping to deliver birthing mothers, or maintaining social stability (Hrdy, 2009). Had genes influencing androphilia also served to engender gentleness and care-giving, they would have conferred a net benefit on kin (Hrdy, 2009). Such persons might have been loved by their families, hence enhancing the survival of offspring of related persons and indirectly transmitting shared genes. As Barron and Hare (2019) suggest, human evolution may have favored a mix of competitive and cooperative forms of personality, with homosexual individuals perhaps being more likely to take nurturing, peace-keeping roles.

Gay alloparents also enhance the overall reproductive success of a family by serving as backup parents. One example is a childless gay aunt helping a sibling raise additional children, thereby increasing the family's overall

reproductive output compared to what the aunt alone might have contributed (Kirkpatrick, 2000). This setup essentially boosts the family's genetic payoff. This is a pattern repeated in the “gay uncle” strategy—sharing resources, protecting, or mentoring that give a child better odds of survival (Kirkpatrick, 2000). Historical accounts of many societies show unmarried aunts and uncles playing vital roles in child care and upbringing, even acting as a type of family safety net during times of distress (Kirkpatrick, 2000). These facilitative roles likely aided in increased survival rates among children, supporting the evolutionary importance of altruism toward kin (Hamilton, 1964).

Finally, it's worth noting that group-level benefits may have played a role (Wilson and Wilson, 2007). Those groups that welcomed homosexual members into caregiving, mediating, or adaptive kinship roles might have done better than those excluding them (Wilson and Wilson, 2007; Hrdy, 2009). Natural selection operates on genes and individuals, but social groups with stronger cohesion and mutual support may have been more resilient (Wilson and Wilson, 2007). Gay people who contributed to group survival—by traveling to help relatives, adopting orphans, or morale lift—may have made their societies more resilient (Hrdy, 2009). In welcoming societies, these behaviors would flourish, explaining why such traits endure (Wilson and Wilson, 2007).

The Evolutionary Logic of Love

Homosexual behavior is not an evolutionary

puzzle, but part of our species' adaptive strategy for survival through cooperation. Biology, anthropology, and zoology all show that non-reproductive individuals can play a role in passing on shared genes by helping relatives (Hrdy, 2009; Silk, 2007). Alloparenting is one explanation for dispelling the apparent paradox of the prevalence of homosexuality (Lee, 1987; Hrdy, 2009). In species that include bonobos, albatrosses, and hominids, non-reproductive individuals contribute significantly to the bringing up of young (Young & VanderWerf, 2014; Silk, 2007). Genetic and twin studies provide evidence for a biological basis for homosexuality, and traits like empathy and low aggression facilitate effective alloparenting (Bailey *et al.*, 2000; Mustanski *et al.*, 2005).

Homosexuality can be seen as an alternative evolutionary strategy that strengthens families and communities. Human child-rearing evolved in a cooperative context—the common phrase “it takes a village” reflects an evolutionary reality (Hrdy, 2009; Kramer, 2010). Within that village, those who help raise children without having their own play a vital role. The kin selection hypothesis reframes evolutionary success as not just producing children, but ensuring the survival of extended kin. This view invites a broader appreciation of the diversity of reproductive strategies in nature, and of the crucial role same-sex oriented individuals have played. Love, care, and cooperation—in all their forms—have always been central to our species' success.

Homosexuality, in this light, is not a mystery

of evolution but a quiet triumph of it—a testament to how love, care, and kinship can shape the survival of generations.

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Artwork by Kah Sin Tang

Haiku for *Callicebus barbarabrownae*

By Alix Atwell Draper

Blur of black and gold
caatinga cattle lowing
distant squawks obscured

For Verreaux's Sifaka

By Janaki Nair

If you long to love a primate,
One who's worthy of your praise,
If you seek a little cutie,
Meet the White Sifaka—sure to amaze.

All sifaka species call one place home,
Madagascar's forests, wild and wide.
But now their numbers fall as they roam—
Over thirty years, a steep decline, worldwide.

They're named for their own calling—
Now how cool is that?
They cry out, "Sifak! Tsibuhak!"
The name just stuck—imagine that!

Sifakas are folivorous, built to browse,
With molars sharp, spiky, and high,
Their teeth can shred through leafy
boughs—
A shearing force no leaf can defy.

A vibrant little leaper,
They cling and leap with flair,
On land they hop so daintily—
Like ballet in the air.

Are they skating in a roller rink,
Arms up for balance, smooth and light,
Or surfing waves that rise and sink,
Their movements flowing, bold and bright?

With eyes so wide and focused tight,
Like tiny moons that softly glow,
Their fur is thick and snowy white—
In Madagascar's trees they go.

These creatures have truly stolen my heart,
I might even tear up—don't ask me why!
So rare, so loud, a world apart—
Better than most... especially the aye-aye.



Artwork by Kah Sin Tang

The Lactase Puzzle: Investigating the Evolution of Lactase Persistence

Joanna Lin, Columbia University

Introduction

Lactase persistence (LP)—the ability to digest the lactose content found in milk into adulthood—is unique to human populations and the subject of extensive research in recent years. Known to be associated with at least five variants, lactase persistence varies widely among human populations, with a monogenic origin in Eurasia (here referring to Europe, Western Asia, and Central Asia, excluding the Middle East) and a primarily polygenic origin in Africa and the Middle East. The lactase persistence (LP) trait is associated with the lactase enzyme, lactase-phlorizin hydrolase (a β -Galactosidase), which is responsible for the hydrolyzation and subsequent breakdown of lactose (Ségurel and Bon, 2017). The lactase enzyme is a small intestinal enzyme, which cleaves lactose into its constituent monosaccharides, glucose and galactose, which are absorbable by mammals (Ingram *et al.*, 2008). In non-human mammals, lactase expression declines after the weaning period, whereas in approximately a third of humans, lactase expression persists throughout adult life (Ségurel and Bon, 2017). The geographical frequency of lactase persistence varies significantly on a global scale, with frequencies ranging from 5% to 100% in different populations (Ségurel and Bon, 2017). The LP trait is also a commonly cited example of niche construction: the process by which an organism constructs and alters

components of its local environmental states, introducing novel selection pressures (Gerbault *et al.*, 2011). Furthermore, the LP trait exhibits one of the strongest selection coefficients within the human genome (Ségurel and Bon, 2017). Despite being one of the most commonly cited examples of niche construction and a highly selected-for trait, the lactase persistence trait remains far from fully understood, with the reasons underlying the selective advantage of lactase persistence remaining highly debated in the field of evolutionary biology.

The global distribution of lactase persistence reveals that lactase non-persistence (LNP) is the most common phenotype in humans at 65%, and lactase persistence is at its highest frequency in Northwestern Europe, with a decreasing cline to the south and east (Ingram *et al.*, 2008). Populations with LP frequencies of over 75% include Northern and Central Europe, and sparse populations in West Africa, East Africa, and the Middle East (Ségurel and Bon, 2017).

This paper aims to understand the reasons for such variation of lactase persistence frequencies across populations, examining the following questions: What is the cause of such physiological differences between human populations? What contributes to the selective pressure for lactase persistence? Can there be more than one factor contributing to such selective pressures? To answer these questions,

this paper will discuss the lactase persistence (LP) phenotype, which is defined by the presence of the lactase enzyme in adults. The lactase non-persistence phenotype (LNP), defined as the absence of the lactase enzyme, differs from the broader term “lactose intolerance,” which refers to any symptoms present after the consumption of lactose, regardless of cause. This paper does not address other lactose intolerance phenotypes such as congenital and infant hypolactasia, milk allergies, or secondary hypolactasia (Ségurel and Bon, 2017). The story of the evolution of lactase persistence remains challenging to piece together for researchers, but with increased ancient and modern-day genetic data accessible for research, we can better comprehend the diffusion of lactase persistence. This paper analyzes the three primary hypotheses proposed to explain positive selection of LP by comparing the research of Holden and Mace (1997) and Gerbault *et al.* (2009). These studies use different methodologies to examine LP diffusion and highlight the complex interplay between genetics, culture, demography, environmental factors, and LP prevalence. Overall, selection for LP appears to have many coexisting explanations specific to different geographical and demographic conditions, contrary to the commonly accepted universal explanation for LP selection. Understanding the persistence and evolution of lactase persistence offers invaluable insight into human biology, as one of the most compelling examples of ongoing human evolution.

Genetic Basis of Lactase Persistence

Within the selective distribution of LP frequency across populations, early researchers found a notable correlation between populations with the cultural practice of drinking milk and the LP phenotype, which led to the hypothesis that the LP trait was subject to strong positive selection pressures, adaptively induced by the environment (Ingram *et al.*, 2008). In the early 1970s, researchers demonstrated that LP is inherited as an autosomal dominant trait, and a decade later, the first evidence was found showing that lactase persistence is both a genetic trait and caused by a cis-acting element of the lactase (LCT) gene (Ingram *et al.*, 2008). The first mutation associated with the LP phenotype was found in 2002 by Enattah *et al.*, who identified the -13.910:C>T mutation found 14 kb upstream of the LCT gene, in intron 13 of the minichromosome maintenance complex component 6 (MCM6) gene. Further studies done by Lewinsky *et al.* in 2005 found that the -13.910:T mutation results in increased transcription through the creation of a new binding site for the octamer-binding protein 1 (Oct-1), a transcription factor that acts through an upregulating pathway that interacts with human hepatocyte nuclear factor 1 α (HNF1 α) to bind to the LCT promoter. In contrast to LNP individuals, the -13.910:T mutation in LP individuals enables LCT expression that is no longer downregulated into adulthood. The -13.910:T mutation is thought to be the causal variant for the LP phenotype found in Eurasia, with correlations between the

distribution of the -13.910:T mutation and the LP phenotype (Ségurel and Bon, 2017). Beyond Eurasia, -13.910:T is also associated with the LP phenotype in India (45% of herders from the North), Mozabites (26%), and other Berber populations from North Africa (Ségurel and Bon, 2017). LP within Eurasia appears to be monogenic, with frequency distributions of LP in broad agreement with the -13.910:T variant (Ségurel and Bon, 2017).

While the -13.910:T mutation was the first mutation found to be associated with the LP phenotype, new variants of intron 13 of MCM6 were determined in subsequent studies, following the surprising discovery of the first regulatory mutation in the MCM6 gene—a gene not within LCT or in its immediate vicinity upstream (Ségurel and Bon, 2017). In the following years, researchers discovered several new mutations associated with the LP phenotype, all of which were in close proximity to -13.910:T. In 2008, Enattah *et al.* found a new -22.018:G>A mutation in intron 9 of MCM6, which was found to be in complete linkage disequilibrium with -13.910, associated with -13.910 in a sample of 236 individuals from four populations (Ségurel and Bon, 2017). -22.018 is thought to interact epistatically with -13.910; *in vitro* studies indicate that -22.018 acts as a weak silencer of enhancer activity driven by -13.910, and there is strong linkage disequilibrium between -13.910 and -22.018 worldwide (Ségurel and Bon, 2017). In East Africa, four newly discovered mutations were found to be associated with LP: -13.907:C>G, -13.915:

T>G, -14.009: T>G, and -14.010: G>C, which are all found in intron 13 of MCM6 (Ségurel and Bon, 2017). LP is commonly associated with five variants: -13.910:T (interacting epistatically with -22.018) in Eurasia, North Africa, and Central Africa; -13.915 in the Middle East; -13.907 (Ethiopia and Sudan); -14.009 (Ethiopia and Sudan); and -14.010 in East Africa (Kenya and Tanzania) and South Africa (Ségurel and Bon, 2017). Although these new genetic candidates for LP phenotypes may appear similar to each other, these mutations appear on distinct haplotype backgrounds, indicating that the derived alleles cannot be markers for a single causal variant (Ségurel and Bon, 2017). This suggests that there must be several independent factors contributing to the lactase persistence trait, illustrating an example of regulatory convergent evolution (Ségurel and Bon, 2017).

Selection Pressures of Lactase Persistence

The selective pressure of lactase persistence was first investigated by Bersaglieri *et al.* (2004), examining the selection signatures of the LCT gene, which showed characteristics of strong and local positive selection. Further analysis indicated that the 13.910:T allele arose 2,188-20,650 years ago with an associated selective pressure of 0.014-0.15 for the United States population (Bersaglieri *et al.*, 2004). In comparison, other research examining the selection coefficients of genes associated with malaria yield values between 0.04 and 0.09, demonstrating the strength of positive selection for LCT in the

human genome (Ségurel and Bon, 2017). Overall, LCT represents one of the strongest signals of recent positive selection documented in the genome, with similar signatures of positive selection found in other populations, all of which exhibit signals of local, recent positive selection (Bersaglieri *et al.*, 2004).

Positive Selection of LP Phenotype – Hypotheses

Given the robust indications of positive selection for LCT in the human genome, the key question becomes: What is driving selection to favor similar regulatory shifts toward selecting for LCT in multiple populations across the globe? What is contributing to such a high selective pressure? A number of hypotheses have been proposed to explain the positive selection for LP; three of the most notable, commonly investigated hypotheses are gene-culture coevolution of LP, calcium absorption, and arid climate.

The gene-culture coevolution of LP was among the first hypotheses proposed to explain the strong selection for the LP trait, as presented by the seminal papers of geographer Frederick Simoons in 1969 and Robert McCracken in 1971. In 1969, Simoons wrote about the strong association between the presence of dairying and the ability to digest milk into adulthood, and correspondingly, the absence of a dairying tradition and the LNP phenotype in adults (Simoons, 1970). McCracken proposed a similar hypothesis in 1971, stating that the food-scarce conditions of pre-industrial society would result in selective pressures favoring the LP trait over

LNP individuals (Wiley, 2018). Simoons and McCracken are credited as the first researchers to posit the gene-culture coevolution hypothesis of LP, suggesting twofold: first, the increased non-specific nutritional benefits of milk under conditions of scarce high-quality food resources, and second, the association between the selective force favoring LP and the cultural development of dairying during the Neolithic Revolution (Wiley, 2018). Crucially, both Simoons and McCracken required an additional condition under which the LP trait would be favored over LNP, emphasizing the necessary scarcity of other non-dairy food resources (Wiley, 2018). Under these conditions, LP individuals would have the ability to consume milk—rich in proteins, fat, minerals, and vitamins—whereas consuming milk for LNP individuals would be costly, with a range of detrimental intestinal symptoms such as diarrhea (Wiley, 2018).

While gene-culture coevolution of LP was among the earliest hypotheses proposed to explain the high selective pressure for the LP trait, many researchers have questioned its validity and proposed additional hypotheses. As opposed to the non-specific nutritional benefits yielded by lactose consumption cited by Simoons and McCracken, in 1973, Flatz and Rotthauwe argued against the general and unspecific benefits of lactose. Flatz and Rotthauwe (1973) proposed that the singularly high proportion of the LP phenotype in northern Europeans is due to the specific selective advantage of increased calcium absorption in environments with low

ultraviolet (UV) irradiation and a low dietary supply of vitamin D. Citing the gene-culture coevolution hypothesis, Flatz and Rotthauwe (1973) argued that carbohydrates were normally not scarce in human societies, rendering the nutritional advantage of lactose consumption purely due to an increased supply of protein and fat. Furthermore, Flatz and Rotthauwe (1973) pointed out that protein and fat can be obtained from milk by LNP individuals, under conditions where small quantities of milk below the amount causing diarrhea and other unfavorable effects are ingested. Alternatively, Flatz and Rotthauwe (1973) argued that the high prevalence of the LP phenotype in northern Europe can be attributed to the enhanced calcium absorption of LP individuals in geographical regions where dietary vitamin D is insufficient. UV irradiation is critical for the physiological conversion of epidermal 7-dehydrocholesterol to pre-vitamin D₃, which forms a biologically active form of vitamin D downstream (Moghaddam, 2020). Vitamin D plays an important role in the body's absorption and retention of calcium and phosphorus, two important components for bone strength and development (Moghaddam, 2020). As Flatz and Rotthauwe (1973) state, low vitamin D levels are commonly linked to diseases associated with deficient calcium supply, including rickets and osteomalacia. However, with increased calcium absorption conferred by the LP phenotype, LP individuals would observe a selective advantage over LNP individuals in their ability to counteract the development of rickets and osteomalacia, decreasing adult morbidity and increasing

reproductive performance (Wiley, 2018).

Other researchers, including Cook and al-Torki (1975) proposed the arid climate hypothesis, explaining the occurrence of the LP phenotype in populations inhabiting specific regions of the world—namely, dry arid environments. Cook and al-Torki (1975) further complicate our understanding of the LP trait, hypothesizing that the positive selective pressures for LP could be episodic and the result of certain severe circumstances including drought, epidemic, and famine. Expanding upon the culture-historical hypothesis, in 1975, Cook and al-Torki pointed out the benefits yielded by the fluid and caloric content of camel milk for desert nomads of the Arabian peninsula and other low-latitude pastoralist groups. In particular, Cook and al-Torki (1975) hypothesized that adults with high levels of lactase enzyme would have increased survival rates due to increased hydration and nutrition in arid environments. Furthermore, with increased selective pressures coupled with episodic events including severe disease outbreaks such as cholera, milk would be a valuable source of uncontaminated water and nutrition (Cook & al-Torki, 1975).

Beyond gene-culture coevolution, calcium absorption, and arid climate—three of the most extensively researched and broadly supported hypotheses—there have historically been several additional hypotheses for the selection of LP, including the dairying barrier and malaria resistance hypotheses. The dairying barrier hypothesis complements a number of other hypotheses, and it was first proposed by

Bloom and Sherman (2005) to provide further stipulations into the environmental conditions required for the selection of the LP phenotype. Bloom and Sherman (2005) hypothesized that the LP phenotype necessitates specific ecological factors; most importantly, conditions that have the capacity to raise milk-producing ungulates—particularly cattle. Bloom and Sherman (2005) add additional context to our understanding of what determines the varying geographical distribution of dairying, stating that specific environments are required for the subsequent evolution of LP. Others have proposed the malaria resistance hypothesis, another hypothesis reliant upon episodic circumstances—malaria outbreaks—to explain the selection of LP. The malaria resistance hypothesis posits that a degree of malaria resistance may be conferred by a milk-based diet, with lactose potentially impairing folate synthesis in malaria infections, leading to a selective advantage of LP (Gerbault *et al.*, 2011; Lokki *et al.*, 2011). These additional hypotheses will not be examined within this paper, but they present complementary hypotheses that highlight the range of possibilities offered by research concerning LP.

Comparative Analysis of Hypotheses for LP Selection: Maximum Likelihood Model

Among the first researchers to analyze the different hypotheses for LP was Holden and Mace (1997), who tested the three prevailing theories for LP selection: gene-culture coevolution, calcium absorption, and arid climate hypotheses. Accounting for phylogenetic confounding

due to shared ancestry of populations with high LP frequencies, Holden and Mace (1997) comparatively analyzed the three hypotheses by regressing the relevant environmental variables (pastoralism, sunshine intensity, and aridity) against the prevalence of the capacity to digest lactose in a sample of different populations worldwide. If unaccounted for, the shared ancestry of sample populations could produce erroneous relationships or obscure authentic associations between variables. By using phylogenetic comparative methods, researchers were able to place their sample populations on a phylogeny and measure change along specific branches of the phylogenetic tree (Holden and Mace, 1997). This allowed researchers to perform regression analysis on independent change events in variables moving across phylogenetic branches, as opposed to simply analyzing the branch tips that correspond to modern phylogenetic traits (Holden and Mace, 1997). Furthermore, Holden and Mace (1997) used a combination of genetic and cultural phylogenies, accounting for genetic and cultural transmission in examining the evolution of LP in populations with high LP frequencies. By accounting for both factors, researchers gained a more comprehensive understanding of the LP phenotype, as the genetic trait of LP and the cultural traits of pastoralism and milk drinking are closely intertwined. Using a maximum-likelihood method indicating the direction of evolutionary change, Holden and Mace (1997) performed their analysis on a dataset composed of a total of 7905 individuals from 62 distinct cultures. The

extensive dataset strengthened their comparative approach, but it also introduced caveats that limit the scope of their conclusions.

However, there were a number of gaps within the dataset, with the loss of samples from non-aboriginal populations of the Americas and Australia in addition to excluded data from other notable populations including the hunter-gatherer Khants of western Siberia, pastoralist Kasakhs, the Romas, and a number of Indian groups (Holden and Mace, 1997). Such populations were either not included within the dataset pulled from the Ethnographic Atlas of 1969 or genetically ambiguous, detracting from the strength of the data and providing future avenues for further, more comprehensive sampling (Holden and Mace, 1997). Holden and Mace (1997) also failed to distinguish between populations that primarily consume processed or derived forms of milk such as cheese and yogurt, which contain low-lactose content, and others that primarily consume fresh milk, which contains high-lactose content. While Holden and Mace (1997) claimed that milk-processing technologies have historically appeared to exist in all pastoralist groups, recent data suggests that consumption practices of milk varied widely between populations. For example, in Central Asia—a location with high numbers of herding populations but low LP frequencies—researchers recently proposed that the Central Asian herders developed alternative adaptive mechanisms to consume milk through fermentation as opposed to genetically, without

the upregulation of the LCT gene (Segurel *et al.*, 2020). In particular, Segurel *et al.* (2020) noted that herding populations in Central Asia primarily drink milk in the form of fermented beverages, making lactose easier to digest, representing a different cultural mechanism of lactose consumption. With our current increased understanding of potential adaptive mechanisms to circumvent the need for the LP phenotype in lactose digestion, as exemplified by the Central Asian herding populations, the research done by Holden and Mace may be cautiously considered, acknowledging that it may not represent the whole picture of selection for the LP trait.

Altogether, using additional quantitative data from the Ethnographic Atlas, Holden and Mace (1997) tested for correlated evolution of LP with the following traits: dependence on pastoralism, solar radiation, dry months per year, and average rainfall. Holden and Mace (1997) used the CAIC program to generate independent contrasts, which are measurements of change at each branch of the phylogenetic tree. Two genetic trees were constructed using different, converging methods of measuring genetic distance, with an F_{ST} linkage tree and a second linkage tree constructed using Nei's method (Holden and Mace, 1997). Accounting for non-genetic factors in LP distributions, a cultural descent of populations was also generated using a language tree, assuming a monophyletic origin of language (Holden and Mace, 1997). Both genetic and linguistic trees were used in the analysis, and to take into consideration

the rapid evolution of language, Holden and Mace (1997) omitted similarities from historical relationships beyond 10,000 years of divergence. This was due to a lack of resolution within deep linguistic historical relationships (Holden and Mace, 1997). Combining genetic and linguistic evidence, Holden and Mace (1997) employed statistical testing using independent and dependent contrasts for both sets of trees, reinforcing the reliability of their modeling by using two converging lines of evidence. The statistical model examined two modes of change: independent change and dependent change (Holden and Mace, 1997). The model of independent change allowed the comparison of two binary variables, measuring each variable's capacity to evolve independently along each branch of the phylogenetic tree (Holden and Mace, 1997). In contrast, the model of dependent change measured the probability of change in one variable dependent on the change in the other variable, examining the relationship between two variables (Holden and Mace, 1997). Using these two modes of statistical analysis, Holden and Mace (1997) tested whether the LP trait—the dependent variable—is statistically more likely to be present under conditions where other variables such as milking are present. The two models of change were fitted to the data using a maximum likelihood technique, statistically determining whether the independent or independent change model yields a better fit to the data (Holden and Mace, 1997). The statistic of interest used a model of

log-likelihood as a measure of goodness of fit, comparing the two models (Holden and Mace, 1997). Using the computer program “DISCRETE”, the researchers used the FST genetic tree paired with linguistic relatedness, specifically pooling data from Beja, Zulu, Ngoni, and Swazi groups, which showed no variation in the relevant variables (Holden and Mace, 1997). The researchers did not pool data from groups outside of the Beja, Zulu, Ngoni, and Swazi groups, which is a lacking area of their analysis, as increased sampling of linguistic relationships could strengthen the validity and reliability of their results with converging evidence. Using the “DISCRETE” program to complete their analysis on both calculated phylogenies (FST linkage tree and modified Nei linkage tree), researchers found that the percentage of reliance on pastoralism explains the greatest amount of variance in LP phenotype frequencies between populations worldwide (Holden and Mace, 1997). However, Holden and Mace (1997) also noted that the distribution of livestock dependence appeared to be more widespread than that of the LP phenotype.

Holden and Mace (1997) also tested their model for populations that traditionally kept livestock but did not practice the tradition of milking their animals, showing that there was no significant difference with the added condition. There was a slight increase in the association between pastoralism and high LP frequency distributions, but the significance level remained the same, accounting for additional scenarios and

reinforcing their results (Holden and Mace, 1997). The maximum likelihood models for independent versus dependent relationships were then compared with a chi-squared test, yielding a significant relationship between high LP frequencies and milking (Holden and Mace, 1997). The maximum likelihood models also provided insight into the direction of evolutionary change, predicting two possible pathways of evolutionary change: LP frequencies dependent on milking, or conversely, milking dependent on pre-existing LP frequencies (Holden and Mace, 1997). Each evolutionary pathway was estimated using the maximum likelihood method, indicating that high LP frequencies did not evolve without the prior presence of milking (Holden and Mace, 1997). The results of Holden and Mace's 1997 analysis support the gene-culture coevolution hypothesis, with high LP frequencies as a broad adaptation to dairying, with no evidence to support the calcium absorption hypothesis of Flatz and Rotthauwe or the arid climate hypothesis of Cook and al-Torki. Using the maximum-likelihood model across a broad worldwide population, Holden and Mace (1997) were among the first researchers to comparatively analyze the different hypotheses for positive LP selection, finding that adult LP evolved in populations following the presence of pastoralism and livestock.

While Holden and Mace's comprehensive and joint analysis of 62 populations remains relevant even in the most recent research covering the evolution of LP, it is critical to note the weaknesses of their methodology and

potential areas of further research. Published in 1997, Holden and Mace's research was extensive in the number of populations covered and hypotheses tested, but the large dataset also poses the risk of overgeneralizations. Holden and Mace's analysis of LP on a worldwide scale meant that smaller-scale relationships between LP frequencies and geographical areas were left potentially unidentified, leading to a broad overgeneralization of the relationship between adult LP and pastoralism. In particular, Holden and Mace noted that the distribution of livestock dependence was more widespread and did not directly align with LP frequency data, showing that there may be smaller-scale factors at play, beyond the general trend of a correlation between pastoralism and global LP distributions. For example, as noted previously, recent research conducted by Segurel *et al.* (2020) examining LP frequency distributions in Central Asia—an exception to the broader trend of pastoralism linked to high LP frequencies—found that Central Asian pastoralists may have adapted culturally as opposed to genetically to consume milk by fermentation, representing an alternative mechanism of milk consumption through derived dairy products. Furthermore, Holden and Mace's analysis presupposed that the hypotheses—the gene-culture coevolution, calcium absorption, and arid climate hypotheses—were mutually exclusive, independently testing each hypothesis to LP distribution data, which presents an additional unaccounted-for scenario of co-existing hypotheses, worth further research.

Further avenues of research, with increased statistical methods and accessibility to modern-day datasets, may fill in the gaps within the dataset used by Holden and Mace, which excluded data from a number of populations (Khants of western Siberia, Kasakhs, the Romas, and a number of Indian groups) and test for both small-scale and global correlations. In a 2009 follow-up published by Mace, she notes that she remains in support of the results of her preliminary study, and if repeated, would still conduct the study using similar methodologies. However, she notes a particular area of weakness in the study: the uncertainty in the tree itself. Newer Bayesian methods of phylogenetic comparison can account for uncertainty in the phylogenetic tree, yielding more accurate and comprehensive insights into the evolutionary path of LP (Mace, 2009). Overall, Holden and Mace's initial research examining genetic and cultural phylogenies to follow the evolution of LP and relevant variables (pastoralism, sunshine intensity, and aridity) contributed substantial insight into our modern understanding of LP, laying the foundation for further research into the cultural, genetic, and geographical selective advantages of the LP phenotype into adulthood.

Comparative Analysis of Hypotheses for LP Selection: Simulation Models

Building off of the research done by Holden and Mace, Gerbault *et al.* likewise examined the positive selection of LP in their 2009 research testing the gene-culture coevolution and

calcium absorption hypotheses on different geographic scales. First, Gerbault *et al.* (2009) examined whether LP frequency distribution patterns could have been shaped by population history and demography—an area of research scarcely addressed within the seminal research done by Holden and Mace. Second, Gerbault *et al.* (2009) used computer simulation models to analyze LP frequencies and pastoralism levels comparatively in Africa and Europe. Contrary to the results of Holden and Mace applied broadly across the globe, Gerbault *et al.* (2009) found that the gene-culture coevolution hypothesis yields a better fit to the observed data in Africa, whereas the calcium absorption hypothesis is more compatible with the observed data in Europe. Drawing from modern population data of 115 worldwide populations (38 African, 34 Asian, 33 European, 7 American, and 3 Oceanian) tested for LP, Gerbault *et al.* (2009) compared these data to pastoralism levels taken from the Ethnographic Atlas and geographic coordinates of the populations of interest.

Gerbault *et al.* (2009) tested the correlation between genetic and geographic variation for selection of the LP-associated allele, assessing whether the LP phenotype could have evolved neutrally at both a global and continental level. Using a dataset of 115 worldwide populations, the researchers determined the correlation coefficients between genetics and geography, finding a weak but significant correlation between LP-associated allele frequencies and latitude on a global scale (Gerbault *et al.*, 2009).

However, it is also noted that this result is the result of sharp contrasts in LP-associated allele frequencies observed in Europe and sub-Saharan Africa, as opposed to a continuum of frequencies (Gerbault *et al.*, 2009). When testing the relationship between LP-associated allele frequencies and longitude, the researchers found a significant correlation that was low and negative (Gerbault *et al.*, 2009). This is reflective of the increased LP phenotype frequencies in Eurasia and Africa, as compared to East Asia. The results of Gerbault *et al.* (2009), yielding significant correlations between LP-associated allele frequencies and both latitude and longitude, suggest that the evolution of the LP-associated alleles was not neutral. On a continental scale, Gerbault *et al.*'s (2009) examination of LP-associated allele frequencies and pastoralism levels yielded no significant correlation between genetic and geographic variation in Asia. In Asia, the distribution of the LP-associated alleles is heterogeneous, with frequencies from 0 to 0.626 in the Near East and India and very low frequencies in East Asia (Gerbault *et al.*, 2009). In Africa, Gerbault *et al.* (2009) found a positive and highly significant correlation between the distribution of LP-associated allele frequencies and latitude, but interestingly, an insignificant correlation with longitude. A linear correlation coefficient between LP-associated allele frequencies and pastoralism levels was also determined for 25 African populations with the available data, yielding a highly significant correlation (Gerbault

et al., 2009). In Africa, this is consistent with the gene-culture coevolution hypothesis: populations with high levels of pastoralism are observed to have correspondingly high LP frequencies.

An additional Samova analysis, an algorithm used to determine genetic boundaries between populations, yielded the same results in favor of the gene-culture coevolution hypothesis. The Samova analysis determines genetic boundaries using geographic location and allelic frequencies as variables (Gerbault *et al.*, 2009). The algorithm sorts populations into groups to maximize the FCT index, representing genetic variation due to differences among different populations (Gerbault *et al.*, 2009). The group structure was tested under 10,000 scenarios to determine the maximal FCT index, revealing two significant boundaries separating three population groups from the rest of the African populations: the Fulani, the Tutsi, and various populations of Beja and the Twaregs separated from other African groups (Gerbault *et al.*, 2009). The additional Samova analysis provides an additional line of converging evidence to support the correlation between LP-associated allele frequencies and pastoralism levels, lending support to the gene-culture coevolution hypothesis in Africa. Like Africa, in Europe, a statistically significant, positive correlation between LP-associated allele frequencies and latitude was found for 33 populations, whereas there was no correlation found with longitude (Gerbault *et al.*, 2009). Within Indo-European populations, the correlation with latitude was

likewise observed to increase with latitude, but the correlation was notably higher compared to other European populations (Gerbault *et al.*, 2009). These results are consistent with selection for the LP-associated allele during the Neolithic Revolution, as Indo-European populations are largely considered to be the descendants of the first Neolithic pastoralists in Europe (Gerbault *et al.*, 2009). The correlation between LP-associated allele frequencies and latitude is also consistent with the calcium absorption hypothesis, where populations with insufficient sunlight in Northern Europe are hypothesized to have compensated by increased levels of lactose consumption.

Furthermore, the researchers performed additional tests using comparisons between LCT variation and other polymorphisms in addition to the SELECTOR software to test whether the observed correlation was truly due to positive natural selection alone rather than population history (Gerbault *et al.*, 2009). Researchers first examined variation in the LCT gene and other polymorphisms, RH and HLA, testing whether the evolution of LP-associated allele frequency was correlated with other independent genes (Gerbault *et al.*, 2009). If LP-associated allele frequencies evolved due to positive selective forces, the expected pattern of variation would yield no correlation with other independent genes (Gerbault *et al.*, 2009). Gerbault *et al.* (2009) determined that there was a highly significant correlation between genetic and geographic variation for RH and HLA at all loci, but a decreased significance between both RH

and HLA with LCT. The decreased significance between both systems (RH and HLA) and LCT suggested that LCT variation was at least partially influenced by evolutionary forces linked to population history, but also influenced largely by natural selection (Gerbault *et al.*, 2009).

The SELECTOR simulation software was used to further investigate this question of how positive selection may have influenced the evolution of the LCT gene in Europe. The algorithm used genetic drift, demographic growth, positive selection, estimated time elapsed since the start of milk consumption, and initial LP-associated allele frequencies (Gerbault *et al.*, 2009). Using these variables, the SELECTOR algorithm simulated the evolution of the positively selected LP-associated allele in a population of diploid individuals over discrete generations (Gerbault *et al.*, 2009). Furthermore, the selective advantage of the LP phenotype was represented by setting the fitness of LP individuals (homozygous or heterozygous) to 1, whereas the fitness associated with the LNP phenotype was set to $1-s$, in which s represented the selection coefficient for each population from Europe and the Near-East (Gerbault *et al.*, 2009). For the different selective hypotheses considered, Gerbault *et al.* (2009) varied the selection coefficient to reflect the nature of either the gene-culture coevolution or calcium absorption hypothesis. To test the gene-culture coevolution hypothesis, Gerbault *et al.* (2009) used a constant selection coefficient for all simulated populations, whereas the calcium

absorption hypothesis used an increasing selection coefficient with the latitude of a given population. An additional third scenario was tested using a combination of two selection coefficients to represent Central European populations belonging to the early Neolithic Linearbandkeramik culture, which hypothesizes that dairy farming was more intensely developed in Central European populations (Gerbault *et al.*, 2009). Multiple independent simulations were run for Indo-European and Near East populations to examine the evolution of the frequency of LP-associated alleles and the best fit for the observed data (Gerbault *et al.*, 2009). Gerbault *et al.* (2009) tested four scenarios: demic diffusion with calcium absorption; demic diffusion with gene-culture coevolution; cultural diffusion with calcium absorption; and cultural diffusion with gene-culture coevolution. Among the tested scenarios, demic diffusion paired with calcium absorption yielded the highest probability of obtaining the observed data at 99.1% (Gerbault *et al.*, 2009). Maximum likelihood tests were used independently across populations to determine the associated selection coefficients that yielded the best fit to observed frequencies with 95% confidence intervals (Gerbault *et al.*, 2009). Using computer simulation models, Gerbault *et al.* (2009) suggest that high LP frequencies in Northern and Western Europe are likely to be the result of an additional selective pressure increasing with latitude, consistent with the calcium absorption hypothesis.

The results of Gerbault *et al.*'s (2009) analysis

suggest that the LP frequencies in Africa are most consistent with the gene-culture coevolution hypothesis, whereas LP frequencies in Europe are best supported by the calcium absorption hypothesis paired with a model of demic diffusion of farming. In Africa, the observed LP frequencies are significantly correlated with pastoralism levels in 25 African populations, which is consistent with the analysis of Holden and Mace, broadly supporting the gene-culture coevolution hypothesis (Gerbault *et al.*, 2009). However, in Europe, a latitudinal gradient of LP-associated allele frequencies was found, and further simulation models were used to determine whether the cause was population history, positive selection on the allele, or both. The simulated scenarios using the SELECTOR algorithm showed that the frequencies of LP in Northwestern Europe cannot be explained by genetic drift alone, suggesting a strong selective force on the LP-associated allele across Northern Europe (Gerbault *et al.*, 2009). While consistent with the calcium absorption hypothesis, Gerbault *et al.* (2009) emphasized that the results do not exclude the potential for other causes for the selective force, such as regional nutritional selection. The cause of the strong selective force latitudinally across Northern Europe remains to be fully clarified, although Gerbault *et al.* (2009) concluded that the calcium absorption hypothesis exhibits the most compatibility with both the LP distribution and simulated demographic effects in Europe. Years after the preliminary research of Holden and Mace, Gerbault *et al.*'s

two-part investigation of LP distributions at both worldwide and continental levels as well as the evolution of such frequencies according to different demographic scenarios presents an alternative perspective on the evolution of LP, where likely scenarios were examined at smaller-scale populations. Contrary to the work of Holden and Mace, Gerbault *et al.* (2009) tested observed distributions for fit across continental levels, observing LP frequencies at both small and large geographical scales. By examining continental scales, Gerbault *et al.* (2009) independently analyzed genetic and geographic variation in Asia, Africa, and Europe. While the research done by Gerbault *et al.* (2009) expanded upon key aspects of Holden and Mace's seminal research by examining the different selection hypotheses at both worldwide and continental scales, there were also a number of limitations within their research methods.

Within their simulation model, Gerbault *et al.* (2009) noted that there were a series of assumptions made using highly limited data prior to the Neolithic transition in Europe. In particular, Gerbault *et al.* (2009) estimated that the LP-associated allele was present but infrequent at a starting percentage of 1%, which could be a source of imprecision within the simulation model. In two other studies, this estimated percentage was higher than 1% (Gerbault *et al.*, 2009). Furthermore, the variable of time elapsed since the start of milk consumption was estimated based on indirect archaeological data, which could introduce further imprecision

(Gerbault *et al.*, 2009). Gerbault *et al.* (2009) also fixed the effective population sizes within their model to 1,000 individuals in the first generation to the carrying capacity of 10,000, which was a significant oversimplification. Gerbault *et al.* (2009) noted that the effective size of populations was undoubtedly higher than 10,000, but the value of 10,000 was chosen to detect selection within the model. Despite the number of simplifications and estimations used within Gerbault *et al.*'s simulation model, the consistency over all sets of simulations is a strong indication of reproducibility. Despite this consistency of their simulations, the researchers notably do not consider the arid climate hypothesis, providing no explanation for its omission, stating that the aim of the work was to test the two main hypotheses (the gene-culture coevolution and calcium absorption hypotheses). However, just as the calcium absorption hypothesis was proposed to explain the LP distributions of Northern European populations, the arid climate hypothesis was also a regionally specific explanation for populations in dry, arid environments—particularly within the Arabian peninsula. Gerbault *et al.* (2009) did not consider aridity as a variable within their analysis, which could present an additional area of oversight, exclusively focusing on geographic variation, pastoralism levels, and genetic patterns of LCT and other independent polymorphisms. As Gerbault *et al.* (2009) suggest within their research, there is likely a combination of hypotheses that may be applicable to different geographic regions and populations, which would

best explain the overall distribution of LP. By omitting the arid climate hypothesis, Gerbault *et al.* (2009) also miss the opportunity to strengthen their findings, failing to assess other potential hypotheses for LP selection. Lastly, further sampling would also enable researchers to better disentangle human population history and natural selection, as Gerbault *et al.* (2009) used a highly limited sampling dataset to represent Asian populations, the world's largest and most diverse continent. There were a mere 34 Asian populations tested for LP within the sampling data, which does not account for the fact that Asia houses the largest population of herders (Gerbault *et al.*, 2009).

This provides a clear direction for future research: including and improving sampling from Asian populations, which also represents a population historically misunderstood, with bafflingly low LP frequencies paired with reliance on pastoralism for millennia (Ségurel and Bon, 2017). The research done by Gerbault *et al.* (2009) underscores the challenges in disentangling the true selection pressures acting on the LCT gene, with regional variations further complicating the narrative. Gerbault *et al.*'s research combining analysis of geographic variation, pastoralism levels, and genetic distributions paired with computer simulation models was a critical step in advancing our understanding of selection of LP, thoroughly examining LP at both continental and global levels. Their findings reveal that while the gene-culture coevolution hypothesis remains in alignment with the distribution of

LP in Africa, the calcium absorption hypothesis, understood with demographic effects, best fits the distribution of LP in Europe Gerbault *et al.*, 2009). Moreover, Gerbault *et al.* (2009) indicate a stronger selective pressure of LP in Northern Europe compared to the South, suggesting potential alternative nuances and explanations for LP selection in Europe. Importantly, the work of Gerbault *et al.* (2009) builds upon prior research, showing that multiple hypotheses may coexist under different geographical and demographic conditions, challenging the notion of a universal explanation for LP selection.

Conclusion: Overarching Findings, Future Avenues of Research, & Modern-Day Relevance

The selective pressure of the LCT gene demonstrates significant and local positive selection, making it one of the most pronounced signals of recent positive selection in the human genome. With the first mutation associated with the LP phenotype found to be -13.910:T linked to Eurasia, North Africa, and Central Africa, subsequent research identified additional mutations on distinct haplotype backgrounds (Ségurel and Bon, 2017). This indicates that there must be several independent factors contributing to the lactase persistence trait across different populations, illustrating a textbook example of regulatory convergent evolution (Ségurel and Bon, 2017). To explain the selective advantage contributing to the pronounced signal of positive selection of the LP trait, researchers proposed three primary hypotheses: the gene-culture evolution, calcium

absorption, and arid climate hypotheses. Holden and Mace were among the first researchers to analyze the different hypotheses for LP in 1997, and they comparatively analyzed the three hypotheses by regressing environmental variables (pastoralism, sunshine intensity, and aridity) against the prevalence of capacity to digest lactose in sample populations at a global scale. Holden and Mace's (1997) comparative analysis of independent and dependent change models found that the dependent model was found to be statistically likely to be present under conditions where other variables such as milking are present. Fitted using a maximum likelihood technique, Holden and Mace (1997) found that the percentage of reliance on pastoralism explains the greatest amount of variance in LP phenotype frequencies between populations worldwide, highlighting pastoralism's pivotal role in LP prevalence. The results of Holden and Mace's analysis provided support for the gene-culture coevolution hypothesis, with high LP frequencies as a broad adaptation to dairying. Expanding upon the groundwork of Holden and Mace, Gerbault *et al.* (2009) also examined the underlying causes for the positive selection of LP in their 2009 research. Testing the gene-culture coevolution and calcium absorption hypothesis at different geographic scales, they found that the evolution of LP was not neutral (Gerbault *et al.*, 2009). At continental scales, they found that the gene-culture coevolution hypothesis was most consistent with the LP frequency distributions correlating strongly with pastoralism levels in

Africa (Gerbault *et al.*, 2009). In Europe, they similarly found a correlation between LP-associated allele frequencies and latitude, and a higher correlation within Indo-European populations (Gerbault *et al.*, 2009). Simulation models corroborated selective pressures favoring LP-associated alleles in Northwestern Europe, supporting the calcium absorption hypothesis (Gerbault *et al.*, 2009). Despite using different methodologies, both studies highlight the challenging interplay between genetics, culture, demography, and environmental factors in shaping LP prevalence, reaching agreement with the gene-culture coevolution hypothesis in African populations.

Within both research teams, challenges in population sampling was a shared weakness, underscoring the need for broader sampling measures and inclusivity, particularly among historically understudied populations such as Central Asian pastoralists. For example, Tibetan herding populations are especially relevant in future research, provided that they represent a misunderstood case of a population historically reliant on pastoralism for millennia with low LP frequencies (Segurel *et al.*, 2020). In Holden and Mace's analysis, they excluded several populations, particularly Asian populations, and failed to consider smaller-scale populations, assuming a uniform cause of LP worldwide. While Gerbault *et al.* (2009) expanded upon this through their examination of smaller-scale populations, they also lacked sampling of Asian herders, with only 34 Asian populations within

their data. With increasing research in these understudied populations, there are also new indications that there may be additional adaptive mechanisms for LP through the gut microbiota, which may further complicate our understanding of LP (Segurel *et al.*, 2020). However, only by conducting further research in these previously understudied populations will we understand the evolution of LP.

Research on LP underscores the complicated relationship between culture, environmental factors, dietary habits, and genetic adaptations, adding to our understanding of the Neolithic Transition and its broader implications on human evolution. These implications extend to population dynamics in Eurasia and Africa, as well as variations in skin pigmentation, particularly in Northern Europe. Moreover, the evolution of LP holds relevance in related fields of public health, medicine, and biology, especially in the modern day where dairy consumption is often advertised, despite the fact that LNP is the most common phenotype in humans at 65% (Ingram *et al.*, 2008). Further research into LP and LNP phenotypes can lead to better solutions for addressing LNP-related health issues, optimizing dietary recommendations, and informing public health policies with a greater understanding of nutrition and digestive health globally. As one of the strongest signals of recent positive selection, the evolution of LP remains the subject of further research, fundamental to understanding of our ongoing evolution in the modern day.

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The Evolution of Earwax Variation in Humans

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If you ever placed your finger into your ear canal and compared the substance inside to that of a friend's, you may realize that earwax texture and color varies across human populations. The ABCC11 gene determines earwax type, with the two phenotypes being either wet or dry (Nakano *et al.*, 2009; Ohashi *et al.*, 2010). What evolutionary advantage—if at all—does wet or dry earwax have in a particular environment? What could have allowed for the selection of a certain type of earwax in humans? This paper aims to answer these questions. First, to provide background, I discuss the general functions of earwax, the physiological and genetic processes that underlie the production of earwax, and the variation of earwax in human populations. Then, I analyze the existing—and oftentimes competing—theories for where, when, why, and how wet and dry earwax evolved. Lastly, I point to avenues of future research, specifically on wet earwax's possible association with diseases like cholesteatoma and breast cancer. If there are links between earwax type and disease, then there could be public health implications that must be addressed by the field of biological anthropology. Overall, this paper highlights how the substance that sits in an individual's ear is important to study; earwax not only protects the inner ear, but it can represent an ancestral history and may even indicate future health risks.

Background

What is earwax?

Earwax, also known as cerumen, naturally occurs in all humans, but the type differs across populations. Many researchers have studied this earwax distribution (McCullough and Giles, 1970; Nakano *et al.*, 2009; Ohashi *et al.*, 2010; Sato *et al.*, 2009; Toyoda *et al.*, 2009; Van Hoewyk, 2012; Yoshiura *et al.*, 2006). The consensus among them is that wet earwax is usually sticky, consists of a yellow or brown pigmentation, and accumulates frequently, while dry earwax is typically flaky, appears as either yellow, gray, or light brown, and is produced less often (Ohashi *et al.*, 2010; Yoshiura *et al.*, 2006). Wet earwax tends to be stickier because it has a greater concentration of lipid and pigment granules than dry earwax does (Shah *et al.*, 2011).

By the time earwax is noticed or felt by an individual, jaw movements have pushed it from the inner to outer ear, causing it to sit at the entrance of the auditory canal (Shah *et al.*, 2011). Stress, anxiety, and fear can cause more earwax to be produced (Shah *et al.*, 2011). Earwax serves many important functions; it acts as a barrier to protect the inner ear from insects, excess water, dust, and infections, and it smooths the external auditory canal (Sevy *et al.*, 2023; Yoshiura *et al.*, 2006). This substance in the ear, along with the dust that it traps, will naturally fall out once it

dries up, but if it is pushed far into the ear canal, such as through using cotton swabs, earwax build up can occur and cause blockage (Shah *et al.*, 2011). This may result in hearing loss, itchiness, odor, and pain (Agrawal and Deshmukh, 2021). Roughly eight million Americans every year undergo medical procedures to remove their earwax as a result of this build up (Sevy *et al.*, 2023). Cleaning out earwax, however, can also damage the inner ear—such as by accidentally tearing part of the auditory canal or the tympanic membrane (Agrawal and Deshmukh, 2021)—if not done properly. Thus, earwax, whether wet or dry, is a natural substance in the ear canal that is beneficial for humans, but must be cared for to prevent build up, blockage, or injury.

Physiological and genetic underpinnings of earwax

Earwax involves both physiological and genetic processes. The ear canal automatically produces earwax through ceruminous glands (Sevy *et al.*, 2023; Yoshiura *et al.*, 2006), which are specialized apocrine glands near hair follicles that lead to the skin's surface (Inaba *et al.*, 1987; Sato *et al.*, 2009). In nonhuman mammals, secretion from the apocrine glands—whether that be in the form of earwax, sweat, or body odor—may act as a pheromone and indicate sexual attraction (Yoshiura *et al.*, 2006). Although secretion from humans' ear apocrine glands do not serve this same function (Yoshiura *et al.*, 2006), it is clear that the glands themselves are not unique to humans.

The primary elements of all types of earwax

include shed skin, keratin, and lipids (Shah *et al.*, 2011). Wet and dry earwax not only differ from each other visually, but they also differ in the quantity and quality of lipids. Inaba *et al.* (1987) found that dry earwax consisted of the following: squalene, sterol esters, wax esters, triglycerides, free fatty acids, and cholesterol. Meanwhile, wet earwax mainly consisted of squalene, triglycerides, free fatty acids, cholesterol, and three other unidentified types of lipids. These unidentified lipids, Inaba *et al.* (1987) concluded, are the causes of the main biological differences between wet and dry earwax.

The ABCC11 gene is pleiotropic, meaning that this gene affects more than one phenotype in humans (Ishikawa *et al.*, 2012). For example, many researchers found that earwax and body odor in humans are both controlled by the ABCC11 gene (Ishikawa *et al.*, 2012; Nakagawa *et al.*, 2018; Nakano *et al.*, 2009; Ohashi *et al.*, 2010; Sato *et al.*, 2009; Toyoda *et al.*, 2009; Van Hoewyk *et al.*, 2012; Yoshiura *et al.*, 2006). The ABCC11 gene is also a type of adenosine triphosphate (ATP)-binding cassette transporter in apocrine glands, resulting in wet earwax to be produced (Ishikawa *et al.*, 2012). A single nucleotide polymorphism in the ABCC11 gene caused a shift from wet to dry earwax and, as a further result, a shift to producing less sweat and less body odor in some human populations (Shah *et al.*, 2011). Additionally, the ABCC11 gene follows the Mendelian inheritance principle; humans with the functioning ABCC11 gene with the GG or GA genotype have wet earwax and increased sweat

and body odor, while humans with the ABCC11 gene variant with the AA genotype have dry earwax and decreased sweat and body odor (Sato *et al.*, 2009; Yoshiura *et al.*, 2006).

Variation of earwax in modern human populations

There is a higher incidence of the functioning ABCC11 gene and thus wet earwax in African and European populations, while the gene variant and dry earwax are more common in East Asian populations (Nakano *et al.*, 2009; Sato *et al.*, 2009). Roughly 80-95% of East Asian populations have the dry, flaky substance (Yoshiura *et al.*, 2006). Meanwhile, populations from Southern Asia, Central Asia, Asia Minor, and the Pacific Islands, as well as Native Americans and the Inuit who have Asian ancestry, all have an intermediate frequency of dry earwax of roughly 30-50% (Yoshiura *et al.*, 2006). Dry earwax in African and European populations appears to be rare, with the frequency ranging from 0-3% (Yoshiura *et al.*, 2006). Although we do not presently know the type of earwax in Neandertals—who resided primarily between western Europe and central Asia (Trinkaus and Howells, 1979)—Goldfield (2019) claims that their Eustachian tubes, which connect the middle ear to the throat and nasal cavity, were flat in adulthood. This differs from modern human morphology (Goldfield, 2019). Because of this flatness, Neandertals had a higher risk for ear infections like “swimmer’s ear,” which is caused by bacteria in the outer ear and can lead to bony growths on their skull if there is repeated exposure to cold water

and poor hygiene (Goldfield, 2019). Thus, it is worth considering whether modern humans’ ear morphology and possible earwax type evolved to act as a barrier against these same infections that may have plagued Neandertals, though future research would be needed to explore this idea.

Theories on the evolution of earwax

Why did wet earwax evolve?

Like many other mammals, it appears that chimpanzees—humans’ closest living relatives—produce cerumen and exhibit wet earwax (Sevy *et al.*, 2023). This yellow, sticky substance is therefore not a unique human trait, but rather may long have been part of the ancestral biological plan of humans. McCullough and Giles (1970), however, propose that wet cerumen evolved repeatedly throughout history to protect humans against otitis, a type of infection in the middle ear, as they migrated into hot and humid climates. The authors propose this because they recognized a correlation between environment and allele frequency in both Mexico’s Yucatec population and in Papua New Guinea’s Papuan population. In Yucatec individuals, McCullough and Giles (1970) found that wet earwax is positively correlated with the mean annual temperature for January and July, precipitation, and dew temperature, but negatively correlated with latitude, suggesting that wet earwax is more prevalent in hot and humid climates that are closer to the equator. They bolstered their theory as they also found low frequencies for dry earwax in populations who had first arrived

in Papua New Guinea. The authors note that the incidence of external otitis is greater in hot, equatorial regions, though humans who have wet earwax tend to experience these infections less often than those who have dry earwax. So, McCullough and Giles (1970) conclude that after humans arrived in Yucatan and Papua New Guinea, wet earwax was evolutionarily selected to protect against infections when living in the tropics, resulting in an increased frequency of the G allele. Their hypothesis implies that wet earwax is not only the ancestral form of earwax in humans, but it has undergone positive natural selection multiple times throughout history as humans migrated.

Where, when, why, and how did dry earwax evolve?

The first issue regarding dry earwax is when this variant evolved. Some researchers suggest that the polymorphism rs17822931—a mutation—in the ABCC11 gene underwent rapid selection in East Asia and subsequently produced dry earwax (Ohashi *et al.*, 2010; Sato *et al.*, 2009). Through modeling and a simulation experiment, Ohashi *et al.* (2010) found that the mutation rs17822931-A was selected 2,006 generations ago, coinciding with the divergence of European and Asian populations. Sato *et al.* (2009) also note that this timing of the mutation explains why European and Asian populations exhibit different frequencies of dry earwax. Additionally, the mutation rs17822931-A may have existed for much longer—since 10,038 generations ago, which is roughly when *Homo sapiens* left Africa—

but remained neutral as no selective pressures had yet acted on it (Ohashi *et al.*, 2010). By outlining the timeline of this mutation, Ohashi *et al.* (2010) conclude that natural selection, rather than some other evolutionary force, must have occurred thousands of years ago to explain the rapid increase in rs17822931-A in East Asian populations.

The second issue is where the variant that controls dry earwax evolved. Though I have stressed throughout this paper that dry earwax is more often found in East Asian populations, Yoshiura *et al.* (2006) found that its frequency is highest in Han Chinese and Korean populations, leading them to believe that the rs17822931-A mutation arose in northeast Eurasia and later spread to other areas. Further, Ohashi *et al.* (2010) point out that the incidence of rs17822931-A appears to be associated with absolute latitude in Asian, Native American, and European populations, meaning that those at higher latitudes tend to have greater frequencies of rs17822931-A than the same populations at lower latitudes. This suggests that this mutation possibly appeared as an adaptation to colder climates (Ohashi *et al.*, 2010).

The final issues to consider are why and how dry earwax evolved—whether directly or indirectly. Individuals with wet earwax, carrying the rs17822931-GG or -GA genotype, may experience axillary osmidrosis, a type of body odor caused by secretions from apocrine glands near the armpit due to an excretory function by the ABCC11 protein (Morioka *et al.*, 2020).

Conversely, those with dry earwax, carrying the rs17822931-AA genotype, experience less sweating and less body odor because the ABCC11 protein is nonfunctional (Nakano *et al.*, 2009; Ohashi *et al.*, 2010; Ishikawa *et al.*, 2012; Yoshiura *et al.*, 2006). Therefore, Yoshiura *et al.* (2006) posit that this dry earwax trait arose not because it was advantageous in itself, but rather as what Gould and Lewontin (1979) call a byproduct or spandrel of another trait favored by natural selection, such as reduced sweating in colder climates. Sweating serves as a heat loss mechanism in warmer climates (Frisancho, 1993), but as humans moved into colder regions where humans would need to retain heat, an individual would be at a disadvantage if they carried the G allele. Excessive sweating results in using energy that could otherwise be put towards a different biological function (Frisancho, 1993). As a result, the ABCC11 variant that reduced sweating and odor could have been selected in colder environments like northeast Asia, with the shift from wet to dry earwax occurring as a natural consequence from this pleiotropic gene.

On the other hand, as mentioned previously, Ohashi *et al.* (2010) propose a cold adaptation hypothesis. Because of the correlation between absolute latitude and dry earwax and because of the rapid increase in dry earwax frequency in East Asian populations, the authors argue that dry earwax evolved as an adaptation. They recognize that dry earwax alone may not have a substantial advantage in a cold environment, but this type of earwax coupled with the other traits in the ABCC11 gene variant, such as less sweating,

together likely increased the fitness of East Asian populations living in colder environments 2,006 generations ago.

The intermediate frequency of dry earwax in Native Americans suggests that another evolutionary mechanism other than natural selection—genetic drift—may have influenced its prevalence. When the ancestors of Native Americans migrated from northeast Asia to the Americas, they carried only a small subset of the genetic variation present in their original population (Halverson and Bolnick, 2008). I believe that this founder effect may have included the dry earwax allele, which may explain its continued presence in modern Native American populations, but further studies are required to confirm this theory. Additionally, these populations may have experienced a genetic bottleneck while living in and moving through Beringia, an area with extreme environmental stressors (Amos and Hoffman, 2010). Because bottleneck events are random and entail a reduction in population size (Allendorf, 1986), it is possible that people with wet earwax were removed from the population, allowing people with dry earwax to further pass on their traits.

Gene flow may be disrupting expected geographic trends of dry earwax in some ancient populations in Hokkaido, Japan (Sato *et al.*, 2009). Among the Ainu, an aboriginal ethnic group of Japan, there is a relatively high frequency of wet earwax (Sato *et al.*, 2009). This is because, as Sato *et al.* (2009) suggest, the Okhotsk people—who originally migrated from East Siberia and had a high frequency of dry earwax—interbred with

the Jomon people, who are ancestral to the Ainu and predominantly carried the wet phenotype. The outcome was a higher proportion of wet earwax in the Ainu, despite their location in East Asia (Sato *et al.*, 2009). This finding challenges Ohashi *et al.*'s (2010) claim that latitude and dry earwax are always positively correlated, demonstrating that gene flow and admixture significantly influence trait distribution. Overall, this shows that the frequencies for both wet and dry earwax in East Asia are complex and layered, but dry earwax nonetheless likely originated in northeast Asia and subsequently spread to other populations throughout the world through multiple evolutionary mechanisms, causing the frequency of dry earwax to vary across groups of humans.

Summary and Conclusions

Though managing earwax in modern society may feel bothersome, especially with the increased use of earphones confronting a person with their potential cerumen build up, earwax serves an important role in protecting a person's inner ear from dust, bugs, and water (Sevy *et al.*, 2023; Yoshiura *et al.*, 2006). Wet earwax seems to have been part of humans' ancestral biology and may have been selected many times throughout history when new populations migrated to hot and humid climates (McCullough and Giles, 1970). Meanwhile, dry earwax appears to be most prevalent in East Asian populations, leading some researchers to propose that the mutation for dry earwax first appeared in northeast Asia around

2,000 generations ago (Ohashi *et al.*, 2010). The puzzling question in the current research is why and how dry earwax evolved. Some researchers believe dry earwax emerged as an adaptation to living in a colder climate (Ohashi *et al.*, 2010), though the Ainu are evidence that not all populations in high latitudes have a high frequency of dry earwax. Others argue that it evolved merely as a byproduct of some other favored trait (Yoshiura *et al.*, 2006). I suggest that the gene variant may have persisted due to genetic drift or gene flow. More research and evidence are needed to confirm any theory regarding the evolution of earwax, as the current literature offers no conclusive findings.

Another avenue for future research is the association between a certain earwax type and diseases, as this may provide insights for why earwax itself evolved and the larger implications of it. Upon finding a similar incidence of the functioning ABCC11 gene and the frequency of cholesteatoma—a type of ear disease that destroys the outer and middle ear and may lead to hearing impairment—in European populations, Nakagawa *et al.* (2018) proposed that there may be a possible association between the two conditions. The frequency of cholesteatoma appears to be low in East Asian populations (Nakagawa *et al.*, 2018), further suggesting that wet earwax and cholesteatoma may be correlated. However, more research is needed to confirm this association and, if true, investigate the mechanisms that allow for this disease to transpire in people with the wet earwax gene.

Wet earwax may also share a correlation

with breast cancer, as Ota *et al.* (2010) and Toyoda *et al.* (2009) found that Japanese women with the wet type tend to have a higher risk of breast cancer than Japanese women with the dry type. Yet, Beesley *et al.* (2010) claim that there is no association between wet earwax and breast cancer in European women. Thus, ancestry and earwax type may be evolutionarily tied to different biological processes, including developing breast cancer, but further research is required before establishing definitive links.

In conclusion, the current research on the evolution of earwax is varied; it is debated whether the mutation for dry earwax evolved for a particular adaptive function, as a necessary byproduct of a different advantageous trait, or for some other reason that has yet to be identified, which is why researchers should continue investigating this topic. There may also be broader implications about earwax that go beyond its functions and its history. Is earwax indicative of other medical conditions? Is it protective against diseases in certain climates? These questions matter not just to the field of biological anthropology but to the public at large, as wet and dry earwax can teach humans about our evolutionary past and about the possible risks that we may face. So, the next time you clean out the sticky or flaky substance in your ear, you should be reminded that there is still so much unknown about the variation among humans.

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